IMVELO SAFE KITCHEN

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Prepared by	OYU Green PTE. LTD.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The project involves the large-scale distribution of improved, fuel-efficient cookstoves (ICS) across rural regions of South Africa. These ICS units are designed to replace the traditional baseline three-stone cookstoves, which are characterized by poor combustion efficiency, high fuel consumption, non-portability, and significant smoke emissions.

The improved cookstoves are portable, consume substantially less fuelwood, and exhibit higher thermal efficiency 45% compared to the baseline technology. In addition to reducing fuel usage, these stoves produce significantly less smoke, thereby reducing indoor air pollution and improving respiratory health among users. The project also offers notable social benefits, including reduced time and effort spent on fuel collection, which alleviates both economic and physical burdens on rural households.

The implementation of the project commenced on 01-May-2025, with the goal of distributing ICS to the Households. Following the distribution phase, a dedicated monitoring period will assess stove performance, collect user feedback, and evaluate improvements in indoor air quality and health outcomes. The project is being implemented in various districts across South Africa, as detailed in Section 1.13 of the main report.

Prior to the project, the vast majority of rural households relied on inefficient three-stone stoves that consumed large quantities of fuelwood and generated heavy smoke. Only 11% of South Africa's population had access to clean cooking technologies[1]. This widespread use of traditional cookstoves contributed to severe household air pollution (HAP), deforestation, and elevated carbon emissions. Alarmingly, HAP is linked to the deaths of approximately 600,000 Africans annually due to related respiratory diseases[2].Cooking with traditional cookstove leads to loss of about 500 million tons of non-renewable wood[3].

By replacing traditional stoves with ICS, the project is expected to significantly reduce greenhouse gas (GHG) emissions through improved combustion efficiency and lower fuel consumption. The estimated annual average GHG emission reduction is 191,930 tCO_2e, with total reductions of 1,343,512 tCO_2e.

[1] https://www.researchgate.net/publication/372430266_Biomass_cookstoves_A_review_of_technical_aspects_and_recent_advances#pfc

[2] https://www.bioenergyforumfact.org/sites/default/files/2018-06/CleanCookingSSA-WEB.pdf

[3] https://winrock.org/wp-content/uploads/2017/09/WinrockCookstoveCombined.pdf

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/	01-May-2025–-30-April-2032	VCS	Validation/verification body name	Seven years
verification	01-May-2025–-30-September-2025	VCS	Validation/verification body name	5 months

1.3 Sectoral Scope and Project Type

Sectoral scope[4]	03. Energy Demand
Project activity type	Type II – Energy Efficiency Improvement Project

This section is not applicable as the Project activity is non AFOLU category

Sectoral scope	N/A
AFOLU project category[5]	N/A
Project activity type	N/A

1.4 Project Eligibility

1.4.1 General eligibility

The primary objective of this project is to reduce greenhouse gas (GHG) emissions by distributing improved cookstoves (ICS) to households, thereby replacing inefficient traditional cookstoves. These traditional devices contribute significantly to GHG emissions, including those identified under the Kyoto Protocol, by burning firewood inefficiently and emitting substantial amounts of smoke and black carbon. In addition to mitigating climate impact, the project also aims to improve indoor air quality and reduce respiratory health issues among users.

The emissions reduction calculation follows a VCS-approved methodology and aligns fully with

the requirements of the Verified Carbon Standard (VCS) Program.

- The project activity is not excluded under Table 1, Section 2.1.3 of the VCS Standard v4.1. Activities related to emission reductions through the replacement of traditional, inefficient

[4] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

[5] *See Appendix 1 of the* VCS Standard

baseline cookstoves with improved, efficient project devices are not listed as excluded activities and therefore fall within the scope of the VCS Program.

- The project has met all requirements related to the pipeline listing process as outlined in the VCS Standard. A complete Project Description (PD) was submitted, and a contract was established with a Validation/Verification Body (VVB) for joint validation and verification services. The project was listed in the Verra pipeline within the required timelines. The stakeholder consultation period was conducted in accordance with Verra's guidelines, and the opening meeting with the VVB was held outside the 30-day public comment window. No public comments were received during this period, and therefore, no revisions were required to the project design. All documentation, including listing representation and validation timelines, have been submitted as per Verra's Registration and Issuance Process, confirming the project's compliance with pipeline listing and validation requirements.
- The project applies the methodology VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves v1.0.7, which is fully eligible under the VCS Program. The project is not part of a larger fragmented initiative and is implemented as an independent activity. The expected annual emission reductions are below 300,000 tCO_2e, confirming that the project does not exceed the scale or capacity limits defined in the applied methodology. Additionally, there are no single clusters of cookstove activities that would breach these thresholds.
- Based on the scope of the activity, methodology applied, adherence to listing and validation deadlines, and project scale, it is confirmed that this cookstove project meets all eligibility criteria under the VCS Program and is compliant with all procedural and technical requirements.

1.4.2 AFOLU project eligibility

The project does not fall under any AFOLU (Agriculture, Forestry, and Other Land Use) categories as defined in Section 3.2 of the VCS Standard. It is therefore classified as a non-AFOLU project. The primary objective is to reduce greenhouse gas (GHG) emissions by addressing energy demand through the distribution of Improved Cookstoves, which replace traditional, less-efficient baseline devices. By enhancing energy efficiency and reducing reliance on firewood, the project targets emission reductions through technological improvements rather than land-use interventions. This clearly positions the project within the scope of non-AFOLU activities under the VCS Program.

1.4.3 Transfer project eligibility

The project does not classify as a 'Transfer Project'. Across the entire lifecycle of the project, it has been developed as per the VCS Standard only and the applicable VCS approved methodology.

1.5 Project Design

☐ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☒ Grouped project

Grouped Project Design

Project Description for Grouped Projects

A grouped project shall be described in a single project description, which shall contain the following

No	Criteria	Justification for project description for grouped projects
1	A delineation of the geographic area(s) within which all project activity instances shall occur. Such area(s) shall be specified by geodetic polygons as set out in Section 3.11.	The project activities are implemented within the geographic boundaries of South Africa, as specified using geodetic polygons in accordance with Section 3.11 of the VCS Standard. The boundaries include underserved tribal communities, and the geographic coordinates of the project areas are clearly provided. This delineation ensures accurate monitoring and prevents overlap with other projects
2	One or more determinations of the baseline for the project activity in accordance with the requirements of the methodology applied to the project	The baseline scenario is the continued use of traditional cookstoves by households, which rely on non-renewable biomass for cooking needs. Surveys conducted in the project areas confirmed the widespread use of inefficient traditional stoves prior to the project. The baseline was determined following the conditions laid out in Section 6 of the applied methodology (VM0050)
3	One or more demonstrations of additionality for the project activity in accordance with the requirements of the methodology applied to the project.	The project demonstrates additionality using the activity method outlined in the applied methodology. There are no government mandates requiring the distribution of improved cookstoves in the project areas, and the project is voluntary. It relies solely on the sale of GHG credits to fund the project activities, confirming that it meets the criteria for regulatory surplus and qualifies under the positive list
4	One or more sets of eligibility criteria for the inclusion of new project activity instances at subsequent verification events.	Criteria include geographic consistency, use of approved technologies, adherence to baseline and additionality conditions. The project database tracks the inclusion of new instances, preventing overlap or double-counting.

5	A description of the central GHG information system and controls associated with the project and its monitoring.	A centralized GHG information system is used to monitor and manage the project. The database is downloaded from the cloud. The cloud is connected to the app developed by OYU Green In-House. Each cookstove has a unique serial number recorded in a mobile application, which tracks distribution, installation, and usage. This system ensures transparency and accurate data collection for monitoring and verification purposes. Beneficiary feedback and field visits during the monitoring period address any issues and enhance data reliability.

1.6 Project Proponent

Organization name	OYU Green PTE. LTD.
Contact person	Mr. Deep Gupta
Title	Director
Address	7 TEMASEK BOULEVARD #12-07 SUNTEC TOWER ONE SINGAPORE (038987)
Telephone	+91-9893758888
Email	deepg@oyugreen.com

1.7 Other Entities Involved in the Project

Organization name	OYU Green Private Limited
Role in the project	Carbon Consultant
Contact person	Mr. Srajan Gupta
Title	Director
Address	Plot No.1, 1st Floor (B), Sector-C, Industrial Area, Govindpura, Bhopal, Madhya Pradesh, India.
Telephone	+91- 989 375 8888
Email	srajang@oyugreen.com

1.8 Ownership

The Project Proponent, OYU Green PTE. LTD., will fully fund the project and retain complete ownership of the project and any potential generation of Verified Carbon Units (VCUs). During the distribution of the Improved Cookstoves and the registration of each participating household, the end user (household or stove owner) will be required to sign a declaration form. This declaration will confirm their voluntary agreement to transfer the ownership rights of any carbon assets generated through the installation and use of the Improved Cookstove to OYU Green Private Limited.

1.9 Project Start Date

Project start date	01-May-2025
Justification	The project start date aligns with VCS Program requirements as it marks the official commencement of activities generating emission reductions, including the distribution of improved cookstoves.

1.10 Project Crediting Period

Crediting period	☒ *Seven years, twice renewable* ☐ *Ten years, fixed* ☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*
Start and end date of first or fixed crediting period	01-May-2025 to 30-April-2032

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
01-May-2025 to 30-April-2026	255,683
01-May-2026 to 30-April-2027	230,754

01-May-2027 to 30-April-2028	208,255
01-May-2028 to 30-April-2029	187,949
01-May-2029 to 30-April-2030	169,625
01-May-2030 to 30-April-2031	153,086
01-May-2031 to 30-April-2032	138,160
Total estimated ERRs during the first or fixed crediting period	1,343,512
Total number of years	7
Average annual ERRs	191,930

1.12 Description of the Project Activity

The project activity involves the distribution of Improved Cookstoves (ICS) to households that continue to rely on inefficient traditional cookstoves. The primary objective is to reduce fuelwood consumption and associated greenhouse gas (GHG) emissions by introducing energy-efficient cooking technologies. The ICS units significantly reduce the quantity of firewood required during cooking through enhanced thermal efficiency and optimized combustion.

The design of the ICS includes higher thermal insulation materials and a forced airflow mechanism that improves heat transfer to cooking pots. This results in quicker cooking times and a direct reduction in firewood usage.

The distribution of cookstoves began in May 2025. Each distributed unit is assigned a unique identification number, which is recorded during the distribution process and included in the beneficiary agreement.

As part of the implementation process, training sessions were conducted for all beneficiaries to ensure proper use and maintenance of the device. The improved cookstove features a combustion chamber that retains heat more effectively than traditional models, achieving a higher combustion rate. An insulation layer further enhances heat retention and energy efficiency, making the cooking process faster and cleaner.

1.13 Project Location

The project boundary for the grouped project is geographical boundary of Republic of South Africa with coordinates 30.5595° S, 22.9375° E[6]

[6] https://www.mapsofworld.com/lat_long/south-africa-lat-long.html

Figure 1: Project Boundary[7]

1.14 Conditions Prior to Project Initiation

The project aims to reduce greenhouse gas (GHG) emissions by distributing Improved Cookstoves (ICS) to rural households in South Africa. By replacing inefficient traditional cookstoves, the project seeks to decrease dependence on non-renewable biomass as a primary source of thermal energy. The introduction of ICS will not only contribute to emissions reduction but also support forest conservation by lowering the demand for firewood.

Prior to project implementation, households in the target region relied heavily on traditional cookstoves that consumed large quantities of firewood, leading to significant emissions and environmental degradation. The baseline scenario reflects this continued use of inefficient, biomass-based cooking practices.

For a more detailed explanation of the baseline scenario, please refer to Section 3.4 (Baseline Scenario) of the project documentation.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

[7] https://www.mapsofworld.com/lat_long/south-africa-lat-long.html

The project is a voluntary initiative undertaken by the project proponent and is not being implemented to comply with any local or national laws or regulatory requirements.

A review of South Africa's environmental laws and regulations has been conducted, including the following:

- The National Energy Act 34 of 2008[8]
- The National Environment Management: Air Quality Act 39 of 2004[9]
- The National Environmental Management Act 107 of 1998[10]
- The Environment Conservation Amendment Act 50 of 2003[11]

No concerns or conflicts have been identified between the improved cookstoves project and the aforementioned laws and regulations.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

8 https://www.gov.za/documents/national-energy-act

9 https://www.gov.za/documents/national-environment-management-air-quality-act

10 https://www.gov.za/documents/national-environmental-management-act

11 https://www.gov.za/documents/environment-conservation-amendment-act

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

Contribution to Sustainable Development

The distribution of Improved Cookstoves (ICS) plays a key role in enhancing access to clean cooking solutions and delivering a wide range of co-benefits. It contributes significantly to public health, gender equality, climate change mitigation, and the reduction of deforestation. This project supports multiple dimensions of sustainable development, as outlined below:

a) Environmental Sustainability

- GHG Emission Reductions:
  The efficient ICS reduces greenhouse gas emissions, particularly carbon dioxide (CO_2), by improving combustion efficiency and decreasing reliance on traditional biomass fuels.
- Forest Conservation:
  Reduced firewood demand helps mitigate deforestation, protect biodiversity and wildlife, and promote carbon sequestration, contributing to long-term ecosystem stability.
- Waste Utilization:
  The use of agricultural biomass residues in the ICS encourages the sustainable use of organic waste, reducing environmental pressure from open burning and land degradation.
- Water Resource Efficiency:
  Improved cookstoves produce less smoke and soot, thereby requiring less water to clean cooking utensils, contributing to better water resource management.

b) Social Sustainability

- Health Improvements:
  Reduced indoor air pollution lowers exposure to harmful pollutants such as PM2.5 and CO, significantly improving respiratory health, especially among women and children.
- Time Savings:
  Time saved from firewood collection can be redirected toward education, childcare, or income-generating activities, especially empowering rural women and youth.
- Enhanced Safety:
  ICS designs reduce the risk of accidental burns and fire-related injuries compared to open fire cooking methods.
- Conflict Reduction:
  By preserving forest resources, the project can reduce intercommunal or interethnic tensions that sometimes arise over limited firewood supplies.

c) Economic Sustainability

- Local Economic Development:
  The project stimulates the local economy through opportunities in distribution, assembly, maintenance, and monitoring of ICS units.
- Fuel Cost Savings:
  Households benefit from reduced expenditure on cooking fuel, enabling more disposable income for other essential needs.

- Labor Efficiency:
  Saved household labor, particularly by women and children, can be diverted to productive economic activities.
- Capacity Building & Employment:
  The project creates employment opportunities and offers skills training, which can enhance future livelihood prospects for community members.

Direct Contribution to SDGs:

SDG	Baseline Scenario	Project Scenario
1 NO POVERTY	Baseline devices led to higher spending due to higher fuel consumption by them	Reduction in the Monetary Expenses towards Fuel and access to Improved Cookstoves. Cooking on the project improved stoves leads to reduced fuel consumption and hence savings on the monetary spendings on fuel
3 GOOD HEALTH AND WELL-BEING	Baseline devices led to health problems as stated by the households in the baseline survey	Improvement in Overall Health due to reduced smoke by the project ICS as compared to less efficient baseline stoves
5 GENDER EQUALITY	Baseline devices consume more time due to requirement of higher fuel collection and low efficiency leads to higher cooking time	Reduced fuel consumption via usage of project devices leads to time savings when collecting fuel. Also, less time is required to cook meals on the project device
7 AFFORDABLE AND CLEAN ENERGY	There is no access to the project devices in the baseline scenario	The distribution of project devices provides households with access to cleaner cooking

	There is no employment generated in the baseline scenario as the project is not active during the same	Employment generated from the project execution such as distribution and maintenance of stoves
	In the Baseline Scenario carbon emissions occur more due to higher demand of fuelwood	The fuel savings in the project scenario vs baseline scenario led to Emission Reductions as less non-renewable biomass is burnt to cook meals

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	1.1	1.1.1 Proportion of population below the international poverty line	Implemented activities to decrease	The implemented Project helps the community by reducing the usage of fuelwood and its expense	The Improved cookstoves reach households and improve the cooking methods and cooking time, hence it reduces the usage of firewood and provides time to work on alternate jobs and it increases the monthly expenses
2)	3.9	3.9.1 Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease	The project activity improves indoor air quality and reduces the emission of fine particulate matter during cooking. This leads to the health outcomes as mentioned in the monitoring survey.	The improved indoor air quality due to the usage of the improved cookstoves results in better health and well-being of the people.
3)	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	Implemented activities to increase	Distribution of ICS to households will reduce the fatiguing work and time spent by women and girls.	The implementation of ICS in the household enables women and girls to spend less time in cooking and spend the remaining time on education and employment purposes.

4)	7.1	7.1.2:Proportion of population with primary reliance on clean fuels and technology	Implemented activities to increase	The Distribution of ICS to the communities where the inefficient cookstoves are replaced and the distributed population will rely on clean technology.	The implementation of the ICS will result in the population using clean technology.
5)	8.3	8.3.1: Proportion of informal employment in total employment, by sector and sex	Implemented activities to increase	Activity: Generating new job opportunities The project will generate new job opportunities through the project	Improved Cookstoves will consume less fuelwood in cooking.
6)	13.2	13..2.2:Total greenhouse gas emissions per year	Implemented activities to decrease	The project activity implemented the ICS to the rural population has significantly reduced the indoor emissions in the project area. Reduction in fuel wood consumption also contributed to forest conservation promotes carbon sequestration	The group project activity achieves an average annual GHG emission reduction of 191,930 tCO2e, with a total reduction of 1,343,512 tCO2e over the entire crediting period

1.19 Additional Information Relevant to the Project

Leakage Management

Leakage emissions related to transportation are estimated using CDM TOOL 12: Project and Leakage Emissions from the Transportation of Freight. According to this tool and as specified in Section 8.2.3 of VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves (Version 1.0), leakage from fuel transportation is only considered when the average transportation distance exceeds 200 km.

If the average transportation distance is less than 200 km, leakage emissions from fuel transport (PEtransp,y) are not included in the calculation, as per the methodology's exclusion criteria. All relevant assumptions and calculations are documented and included in the Emission Reduction (ER) calculation sheet.

Commercially Sensitive Information

No commercially sensitive information has been hidden from this public version of the project description.

Further Information

No further information

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	In the process of identifying the stakeholders the PP engaged with the community leaders, local leaders, NGOs, and other government representatives to understand the needs of the vulnerable group. The key stakeholders identified local households, women, and girls.
Legal or customary tenure/access rights	The project ensures respect for legal and customary rights to territories and resources held by stakeholders,

	Indigenous Peoples (IPs), and local communities (LCs). These rights remain safeguarded, with no recorded conflicts. Open communication and grievance mechanisms address concerns related to access or tenure rights.
Stakeholder diversity and changes over time	The project reflects significant changes in the group by social, economic, and cultural diversity. Rural households, particularly women and girls are mostly affected by traditional cooking practices. Economically, these households often belong to low-income brackets, where cost savings from reduced firewood use can significantly enhance their livelihoods by economic wise and social attributes. Panchayat leaders and local officials bring governance and administrative perspectives, while NGOs contribute expertise in rural development and health awareness.
Expected changes in well-being	Under the baseline scenario, the stakeholders, households particularly involved in the traditional cooking methods would have an adverse effect on the women and children in the household. The reliance on inefficient stoves would demand significant time and effort for firewood collection, perpetuating drudgery and reducing opportunities for education and income generation. Without intervention, these adverse effects would likely intensify, further eroding community well-being and environmental sustainability.
Location of stakeholders	The program is planned to implement in various districts of South Africa. The project main aim is to impact rural communities and indigenous peoples. These stakeholders benefit directly from reduced indoor air pollution, lower firewood dependency, and improved health and economic savings. Customary rights holders also experience positive impacts, including forest preservation and enhanced livelihoods. Communities outside the project area may benefit from knowledge-sharing and sustainable practices.
Location of resources	The project aim is deployment of new ICS to the participating households and does not involve resources owned by the stakeholders.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	It will be Updated after the Stakeholder Consultation
Stakeholder engagement process	The stakeholder engagement process will be conducted in a culturally appropriate manner by distributing invitation posters and sending official invitations to community leaders, group representatives, and government officials at least 30 days prior to the scheduled meetings. These meetings will be organized with careful attention to language and gender sensitivity to promote inclusive participation. During the sessions, feedback and opinions will be gathered from participants through open discussions and written forms. All feedback received will be documented and taken into account to enhance future project activities and stakeholder engagement processes.
Consultation outcome	The Project Proponent (PP) will communicate the benefits of adopting Improved Cookstoves (ICS), emphasizing cleaner and more efficient cooking practices. The ICS units will significantly reduce smoke emissions, leading to improved health and well-being, particularly for women and children. They will also lower fuelwood consumption, thereby supporting environmental sustainability and contributing to the reduction of greenhouse gas emissions. Additionally, the use of ICS will reduce the time and physical burden associated with collecting fuelwood. During stakeholder engagements, discussions will include the project's alignment with national laws and regulations, including adherence to workers' rights and safety standards. The principles of Free, Prior, and Informed Consent (FPIC) will be upheld to ensure that stakeholders clearly understand the project's design, implementation, associated risks, and anticipated benefits. The validation and verification process under the Verified Carbon Standard (VCS) will also be explained. Overall, the project will aim to enhance community well-being and contribute to broader climate change mitigation efforts.

Ongoing communication	Ongoing communication with stakeholders will be maintained through regular meetings and consultations. The Project Proponent will provide clear contact details and a physical address to ensure stakeholders can easily reach out with grievances, feedback, or suggestions. This accessible communication channel will enable the timely resolution of concerns and support the continuous improvement of project design and implementation.
Stakeholder input	During the stakeholder consultation meeting, the feedback and comments received from participants are expected to be largely positive and supportive of the project. It is anticipated that no significant concerns or objections will be raised that would necessitate changes to the project design. Therefore, based on the input received, no updates to the project design will be required at this stage. However, the Project Proponent will continue to consider ongoing feedback to ensure the project remains beneficial to the community and all stakeholders throughout its implementation.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Consent for the project will be obtained through transparent stakeholder consultations involving community leaders, Indigenous Peoples (IPs), local communities (LCs), and customary rights holders. It is anticipated that no conflicts will be reported, and the project is not expected to influence or exacerbate any existing unresolved issues.
Outcome of FPIC	The project will involve the construction and distribution of energy-efficient improved cookstoves and will not require any land acquisition. Installation of these cookstoves will take place at the residences of the project beneficiaries. The field team will visit beneficiary households only with the consent of the homeowners. The Project Proponent will ensure that all property rights are acknowledged, respected, and upheld throughout the implementation. Additionally, the project will not involve any involuntary relocation, land

	encroachment, or forced physical or economic displacement of individuals.

2.1.4 Grievance Redress Procedure

Development process	The project proponent maintained a stakeholder register at the project office to document feedback and grievances. A dedicated contact number was provided for beneficiaries to raise complaints or concerns. The field team is responsible for receiving, recording, and addressing grievances in a timely manner, ensuring culturally appropriate conflict resolution methods are applied to resolve issues fairly and transparently.
Grievance redress procedure	The grievance redress procedures were developed in consultation with stakeholders to ensure transparency and accessibility. A dedicated contact number and grievance register are maintained at the project office for beneficiaries to submit complaints. Grievances can be raised verbally or in writing and are addressed by the field team within a reasonable time frame. As there was no grievance raised, no resolution was taken into account.

Grievances received	Resolution and outcome
There were no grievances raised during the current monitoring period.	As there was no grievance raised, no resolution was taken into account.

2.1.5 Public Comments

Comments received	Actions taken
This section will be completed after the public commenting period has concluded.	

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project proponent has assembled a team with prior experience in similar project activities, including cookstove projects, rural development, and GHG emission reduction projects. The team is well-versed in designing and implementing projects, conducting stakeholder consultations, and ensuring compliance with international standards. Additionally, the project proponent has hired well-experienced individuals to strengthen the team and support project implementation, monitoring, and reporting activities.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No Risk Identified	The project focuses on providing improved cookstoves that reduce air pollution and fuelwood collection time. Mitigation measures include continuous monitoring of environmental impacts, community training, and the promotion of cleaner cooking practices.
Risks to stakeholder participation	No Risk Identified	The project ensures continuous stakeholder engagement through community meetings, awareness programs, and training sessions. Clear communication channels are established to maintain transparency and encourage participation throughout the project duration.
Working conditions	No Risk Identified	The project prioritizes the safety of women and girls by reducing their exposure to indoor air pollution and minimizing the need for long-distance fuelwood collection. Additionally, community

		awareness programs and regular monitoring are conducted to ensure their safety and well-being throughout the project implementation.
Safety of women and girls	No Risk Identified	The project ensures the safety of minority and marginalized groups, including children, by promoting inclusive participation, reducing exposure to indoor air pollution, and minimizing the need for fuelwood collection. Continuous monitoring and community engagement activities are conducted to safeguard the well-being of these groups.
Safety of minority and marginalized groups, including children	No Risk Identified	The safety of minority and marginalized groups, including children, as the project follows inclusive practices and ensures equal protection and opportunities for all participants.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No Risk Identified	The project promotes the use of improved cookstoves that reduce air pollutants such as particulate matter and carbon monoxide compared to traditional cookstoves. No hazardous materials, chemical pesticides, or fertilizers are used. Continuous monitoring ensures minimal environmental impact, and proper waste management practices are followed during

		distribution and maintenance activities.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified[12]	Mitigation or preventative measure(s) taken
Discrimination	No Risk Identified	The Project ensures inclusivity and equal participation of all stakeholders, preventing any form of discrimination based on gender, ethnicity, or socio-economic status.
Sexual harassment	No Risk Identified	The project maintains a zero-tolerance policy for sexual harassment, implementing strict guidelines, training sessions, and grievance mechanisms to ensure a safe and respectful environment for all participants.
Equal pay for equal work	No Risk Identified	The project proponent make sure that the gender participation in the project are equal and they are equally paid during the program
Gender equity in labor and work	No Risk Identified	The project fosters gender equity by ensuring equal opportunities, fair wages, and access to training and leadership roles for all genders.
Forced labor	No Risk Identified	There are no individuals who are subjected to forced labor, and all work is voluntary with fair compensation and proper working conditions.

[12] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Child labor	No Risk Identified	The project strictly prohibits child labor, ensuring that all workers meet the legal age requirements and work in safe conditions.
Human trafficking	No Risk Identified	The project activities do not involve or promote any form of human trafficking. Strict adherence to national and international labor standards is ensured, and monitoring is conducted to prevent any such risks.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No Risk Identified	The project involves the distribution of Improved Cookstoves (ICS) and does not impact the rights, lands, or resources of Indigenous Peoples (IPs), Local Communities (LCs), or customary rights holders. All activities respect human rights and comply with international standards, including the UN Declaration on the Rights of Indigenous Peoples and ILO Convention 169. Continuous monitoring will ensure rights are recognized, respected, and protected.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No Risk Identified	The project focuses solely on distributing Improved Cookstoves (ICS) and does not involve any activities that affect Indigenous Peoples' rights or cultural heritage. The project does not involve land use, relocation, or interfere with any cultural or sacred sites. All activities respect cultural heritage and uphold the rights of Indigenous Peoples and Local Communities.

	Regular monitoring ensures protection of cultural heritage throughout the project.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No Risk Identified	The project only involves the distribution of Improved Cookstoves (ICS) to participating households and does not require land acquisition or access to property or resources. There is no impact on legal or customary tenure rights or property rights of Indigenous Peoples (IPs), Local Communities (LCs), or customary rights holders. The property rights of all stakeholders remain fully protected throughout the project.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	The process was designed keeping in mind the benefits to accrue to all the stakeholders involved. Users were provided with the improved cookstove free of cost to reduce the consumption of woody biomass. The use of the project device created a healthier environment by reducing indoor air pollution caused by traditional cookstoves, ensuring both environmental and health benefits to the community.
Summary of the benefit sharing plan	The benefit-sharing plan ensured free distribution of improved cookstoves to all stakeholders, reducing fuelwood usage and indoor air pollution. It also created local employment opportunities for distribution and maintenance. No renegotiation or opt-outs occurred during the monitoring period. The plan remains unchanged with continuous stakeholder engagement.
Approval and dissemination of benefit sharing plan	The benefit-sharing agreement was discussed and agreed upon during stakeholder consultations with the affected communities. It will be shared in a culturally appropriate manner through community meetings and consultations with local leaders. A copy of the agreement is available at

	the project office and accessible to stakeholders upon request.
Benefit sharing during the monitoring period	The benefit-sharing plan was implemented by distributing improved cookstoves free of cost to the targeted households. This reduced their dependency on fuelwood, lowering household expenses and indoor air pollution. As a result, communities experienced health benefits, saved time on fuel collection, and contributed to environmental sustainability. All outcomes were shared transparently with stakeholders.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No Risk Identified	The project positively impacts biodiversity and ecosystems by reducing deforestation through decreased firewood consumption, thereby preserving habitats and maintaining ecological balance
Soil degradation and soil erosion	No Risk Identified	The project does not contribute to soil degradation or erosion, maintaining the integrity and fertility of the land in the project area.
Water consumption and stress	No Risk Identified	The project does not increase water consumption or contribute to water stress in the project area, ensuring sustainable resource use

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

Species and habitat	The project activity does not affect the species or habitat. Apart from traditional cooking, the project promotes conserving the forest.
Areas needed for habitat connectivity	The project did not impact areas essential for habitat connectivity, as activities were confined to households and

	did not involve land-use changes or deforestation. Reducing firewood demand indirectly supported forest conservation and ecological linkages.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No Risk Identified	The project operates within household environments, with no impact on habitats for rare, threatened, or endangered species, as confirmed through monitoring and consultations
Areas for habitat connectivity	No Risk Identified	The project does not interfere with areas required for habitat connectivity, ensuring no disruption to wildlife movement or ecological linkages during its implementation.

2.4.2 Introduction of species

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A			

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No Risk Identified	The project avoids land-use changes and biodiversity disruptions by focusing solely on distributing improved cookstoves. Regular monitoring ensures

2.4.3 Ecosystem conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No Risk Identified	The Project adheres to sustainable land management practices, and no invasive species have been introduced.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0050	Energy Efficiency and Fuel-Switch Measures in Cookstoves	1.0
Tool	Tool 30	Calculation of the fraction of non-renewable biomass	4.0
Guideline	CDM-EB50-A30-STAN	Standard for sampling and surveys for CDM project activities and programme of activities	4.0
Standard, 09	CDM-EB50-A30 STAN	Sampling and surveys for CDM project activities and programme of activities	9.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0050- ENERGY EFFICIENCY AND FUEL-	The project activity corresponds to: a) Replacement of non-renewable biomass (e.g.,	The project distributes improved cookstoves that reduce fuel consumption

SWITCH MEASURES IN COOKSTOVES(Version 1.0)	firewood, charcoal)-fired cookstoves with any of the following: i) More efficient project devices that use the same fuel as in the baseline; ii)Efficient project devices fired by renewable biomass or bioethanol; iii)Efficient project devices fired by liquefied petroleum gas (LPG); or iv) Electric-powered project devices.	while maintaining the same fuel type, ensuring compatibility with baseline conditions.
…	Project devices are used in households, communities, institutions, or small or medium enterprises (SMEs),collectively referred to in this methodology as the "target population."	The main objective of the project is to provide improved cookstoves (ICS) to households, replacing traditional cookstoves. The target population for the project consists of households
	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%.	The Project cookstoves meet or exceed the 25% thermal efficiency threshold, as verified through laboratory and field testing
	The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these	Awareness campaigns, subsidies, and financing options are integrated to encourage users to switch from traditional stoves to the improved cookstoves.

	mechanisms in the project description.	
	Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device.	The project proponent (PP) monitored households during the monitoring period to address any issues with the improved cookstoves (ICS). Any problems reported were resolved by the team
	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographic coordinates, complete address information).	The ICS provided by the PP has unique identification number and it also registered through the application with the geographics coordinates and complete information are stored through the application.
	National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented.	The project complies with all national and sub-national policies governing clean cooking solutions and energy efficiency standards.
	Where project activities reduce emissions from non-	The project reduces firewood usage by distributing

	renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description.	improved cookstoves (ICS). Additionally, KML files are provided to define the project boundary and document the geographic locations of the activity.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Delivery of thermal energy (fuelwood, majorly on inefficient cookstoves)	CO_2	Yes	Major Source of Emission
		CH_4	No	Important source of emission
		N_2O	No	Important source of emission
		Other	NA	-
Project	Delivery of thermal energy (fuelwood using efficient cookstoves for cooking)	CO_2	Yes	Major Source of Emission
		CH_4	No	Important source of emission
		N_2O	No	Important source of emission
		Other	NA	-

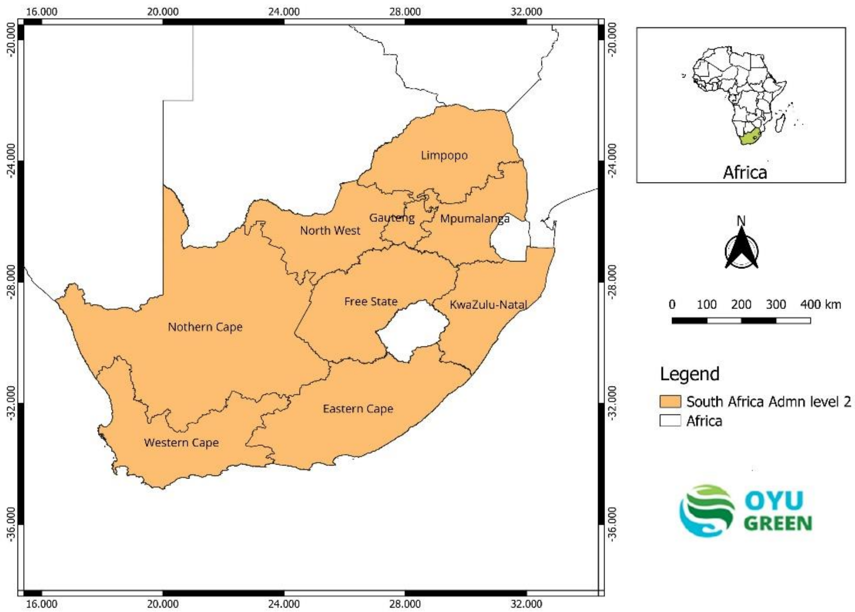

Figure 2: Geographical Boundary of the South Africa

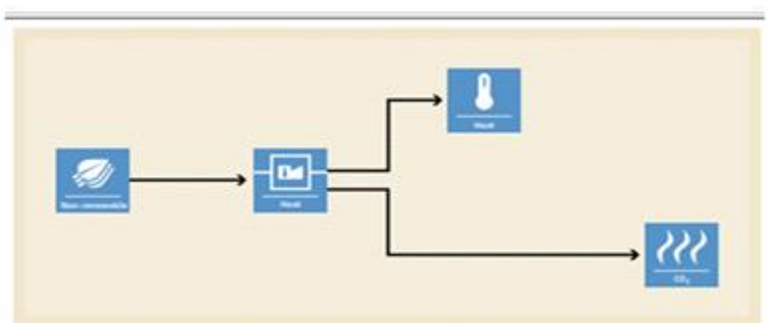

Figure 3: Baseline Scenerio Diagram

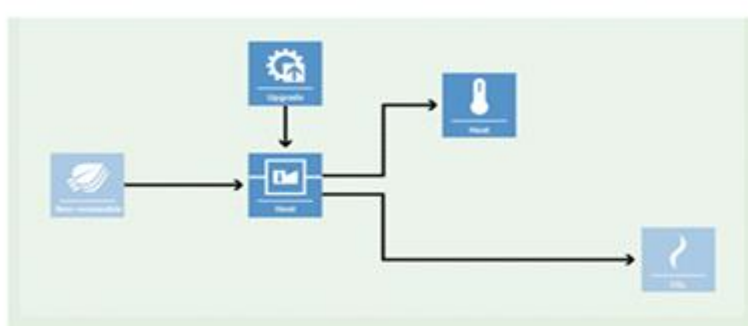

Figure 4: Project Scenario Diagram

3.4 Baseline Scenario

In accordance with VM0050 v1.0, the baseline scenario has been established using a three-step approach: identifying alternative scenarios, evaluating regulatory requirements, and assessing financial, institutional, and informational barriers. The continued use of traditional cookstoves, particularly the three-stone fire, is the most common and realistic alternative in the project area.

These stoves are inefficient, consume large amounts of firewood, and emit high levels of CO_2, CH_4, and particulate matter. A baseline survey will be conducted to confirm that most households rely on such stoves, which lack combustion control and ventilation features. There are no binding national or local policies requiring improved cookstoves, and existing initiatives are voluntary. Without carbon finance, the uptake of improved stoves is unlikely. Therefore, the continued use of three-stone fires is considered the most plausible and conservative baseline. This approach aligns with methodological guidance and real-world conditions.

3.5 Additionality

According to the Section 7 of the applied methodology, additionality will be demonstrated based on the following steps,

Step 1: Regulatory Surplus

There are no mandatory government policies or regulations in the host country requiring the distribution of improved cookstoves. The project operates on a purely voluntary basis, with households choosing to participate without any legal obligation. Since there is no enforced law or statute mandating the use of efficient cookstoves, the project is considered additional.

Step 2: Positive List

The project meets all the applicability conditions defined in the methodology. Improved cookstoves (ICS) are distributed free of cost to households, and the project does not receive funding from government programs or multilateral sources. The sole revenue stream for the project is the sale of GHG credits, making it reliant on carbon finance for implementation.

Step 3: Project Method

If any project activity instance does not meet the positive list criteria (e.g., stoves are not provided for free or the project has other revenue sources), additionality will be demonstrated using an investment analysis as per the VCS Tool Additionality Assessment (VT0008). This analysis will prove that the project is either not the most financially attractive option or not economically viable without carbon credit revenue.

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

3.5.2 Additionality Methods

This section will be updated later

3.6 Methodology Deviations

Not applicable. There are no deviations from the methodology in this project.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

This section will be updated later.

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

This section will be Updated later.

5.2 Project Emissions

This section will be updated later.

5.3 Leakage Emissions

This section will be updated later.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

This Section will be updated later.

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
01-May-2025 to 31-December-2025						
01-Januray-2026 to 31-December-2026						
01-Januray-2027 to 31-December-2027						
01-Januray-2028 to 31-December-2028						
01-Januray-2029 to 31-December-2029						
01-Januray-2030 to 31-December-2030						
01-Januray-2031 to 31-December-2031						
01-Januray-2032 to 30-April-2032						

Total						

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{b,i,CO2}$ $EF_{p,j,CO2}$
Data unit	t CO2/TJ
Description	CO2 emission factor for fuel used by baseline device type i in the baseline scenario CO2 emission factor for fuel used by project device type j in the project scenario
Source of data	Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	112
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories.
Purpose of data	Calculation of baseline and project emissions
Comments	--

Data / Parameter	EFb,i,nonCO2 EFp,j,nonCO2
Data unit	t CO2e/TJ
Description	Non-CO2 emission factor for fuel used by baseline device type i in the baseline scenario Non-CO2 emission factor for fuel used by project device type j in the project scenario

Source of data	Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	9.46
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	NCVb,i NCVp,j
Data unit	TJ/tonne or TJ/m3
Description	Net calorific value of baseline fuel used by baseline device type i Net calorific value of project fuel used by project device type j
Source of data	Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	The values for wood in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Purpose of data	Calculation of baseline emissions
Comments	---

Data / Parameter	η old,avg
Data unit	Fraction
Description	Weighted average efficiency of baseline devices that are replaced by project devices
Source of data	Baseline Survey
Value applied	15%

Justification of choice of data or description of measurement methods and procedures applied	Default Value Taken
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	fNRB,y
Data unit	Fraction
Description	Fraction of woody biomass that is established to be non-renewable used by device in year y
Source of data	Calculated based on CDM TOOL30 and applying an uncertainty discount of 26%
Value applied	0.64
Justification of choice of data or description of measurement methods and procedures applied	This parameter is determined ex-ante. As per Tool 30 Version 4, Detailed calculation will be shared with the VVB.
Purpose of data	Calculation of baseline emissions
Comments	-

6.2 Data and Parameters Monitored

Data / Parameter	N y,i,j
Data unit	Number
Description	Number of commissioned project devices of type j from batch k in year y
Source of data	Monitoring
Description of measurement methods and procedures applied	Number of devices distributed under the project activity

Frequency of monitoring/recording	At least once every two years
Value applied:	XXXXX
Monitoring equipment	Monitoring Survey
QA/QC procedures applied	--
Purpose of data	Calculation of Baseline and Project Emission
Calculation method	--
Comments	--

Data / Parameter	n j,k,y
Data unit	Fraction
Description	The proportion of commissioned project devices of type j from batch k that are still being used regularly in year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and program of activities.
Frequency of monitoring/recording	Annually
Value applied	95%
Monitoring equipment	Monitoring Survey
QA/QC procedures to be applied	Not Applicable
Purpose of data	Calculation of baseline and project emissions
Calculation method	--

Comments	--

Data / Parameter	$\eta_{new,avg,y}$
Data unit	Fraction
Description	Weighted average efficiency of project devices in year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Manufacturer specification or Certification by a national standards body or an appropriate certifying agent recognized by that body.
Frequency of monitoring/recording	Annually
Value applied	XX
Monitoring equipment	Not Applicable
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	--

Data / Parameter	$BC_{p,j,k,y}$
Data unit	Tonnes or m3
Description	Average quantity of fuel used by project device during year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Manufacturer specification or Certification by a national standards body or an appropriate certifying agent recognized by that body
Frequency of monitoring/recording	Annually
Value applied	XX

Monitoring equipment	Monitoring Survey
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	--

Data / Parameter	BCb,i,y
Data unit	Tonnes or m3
Description	Fuel used per baseline device type I during year y
Source of data	Value obtained from the baseline survey
Description of measurement methods and procedures to be applied	The value of BCb,i,y (baseline fuel consumption per household per year) is determined through baseline surveys and measurement campaigns conducted in the project area. The surveys involve direct observation and data collection from representative households using traditional cookstoves. Sampling follows the most recent version of the CDM Standard for Sampling and Surveys for Project Activities and Programmes of Activities, ensuring a confidence level of 90% and precision of ±10%. Household-level data is collected through interviews, physical fuel measurements, and validated with historical records or similar projects in the region
Frequency of monitoring/recording	Biennially or Every two years
Value applied	XX
Monitoring equipment	N/A
QA/QC procedures to be applied	The monitoring campaign aims to achieve at least 90% confidence and ±10% precision for the target parameter of average daily fuel consumption per adult equivalent, as outlined in the CDM Standard.

Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

6.3 Monitoring Plan

The Monitoring Plan that will be applied in this project consists of several core components that are designed to ensure that the Project Proponent (PP) and/or implementing team obtain high-quality, unbiased, and reliable data regarding the project's implementation and results. These elements are essential for the accurate calculation of Verified Carbon Units (VCUs) in accordance with Methodology VM0050 (Version 1.0).

Key Monitoring Elements:

- Structured Data Collection Procedures
- Cookstove Distribution and Monitoring Database
- Random Spot Verification of Installed ICS Units
- Survey-Based Sampling Plan for Monitoring
- Data Quality, Consistency, and Duplication Control
- Periodic Monitoring and Reporting

The Below flowchart is developed to outline the roles and responsibilities of the PP during both implementation and monitoring phases.

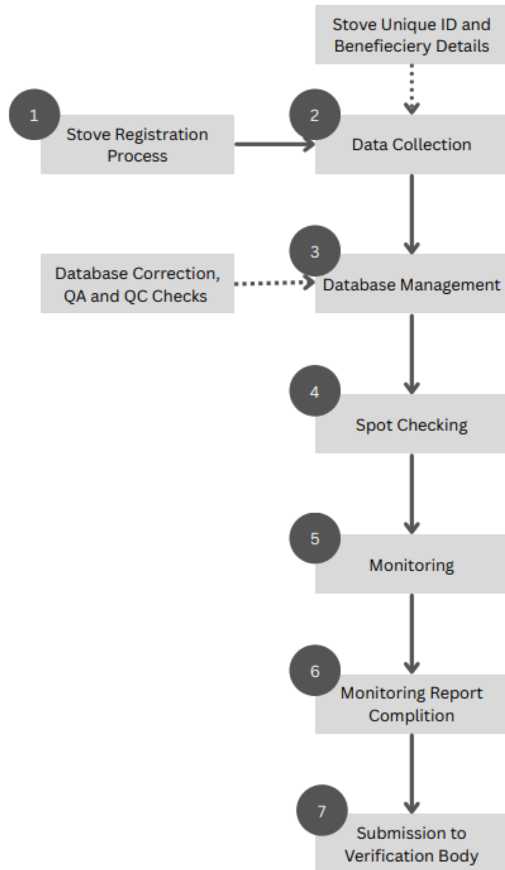

Figure 6: Flowchart of Responsibilities

User Enrolment and Beneficiary Data Collection

A proprietary, cloud-based mobile application will be used to capture beneficiary information at the time of cookstove distribution. Key data points such as name, address, phone number, government-issued ID, family size, and date of distribution will be collected. Each household will be assigned a **unique identification number**, and photographs will be taken to confirm receipt of the stove.

The mobile application will allow real-time data access and will store the following information:

- Full name of the recipient
- Residential address
- Contact number
- Government ID reference
- Total number of household members
- Date of ICS distribution

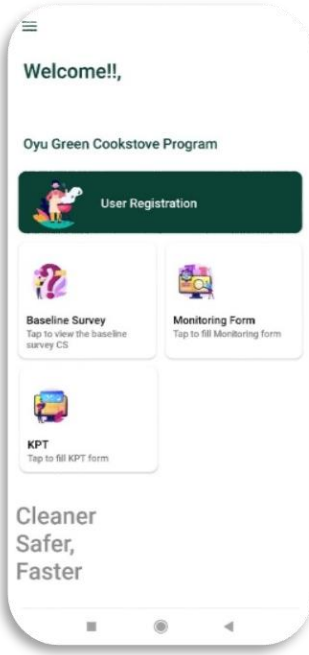

Figure 7:Mobile Application for User Registration

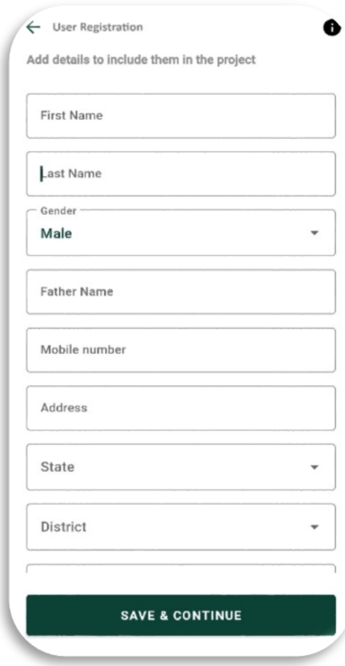

Figure 8: User Registration Window

Screenshots of this mobile platform will be included in the report to demonstrate its dual function—data collection during distribution and usage monitoring in later phases.

Data Management System

Trained field personnel will input beneficiary and stove data into the cloud database via the application. This may occur either through manual entry or automated synchronization. All entries will be subject to consistency checks. Duplicate entries will be flagged for investigation and either corrected or removed to ensure data integrity.

Spot Checks and Data Verification

The project team will conduct random verification checks on selected units within the database. Field staff will perform surprise household visits to validate stove usage and ensure alignment with database records. Any unused or uninstalled cookstoves identified during these visits will be excluded from emissions reduction calculations.

Internal Quality Assurance and Quality Control

The PP will implement internal QA/QC protocols to ensure the monitoring data meets required precision levels. This process will account for potential non-responses and allow for removal of statistical outliers. Regular audits of the database will be conducted to identify discrepancies or missing data. Data validation will include consistency checks, completeness verification, and cross-referencing against physical evidence collected during monitoring.

Sampling Approach

The project will use a **representative sampling method** aligned with the "Guidelines for Sampling and Surveys for CDM Project Activities and Programme Activities (Version 4.0)." Due to the wide-scale distribution of ICS units, a full census will not be economically feasible; therefore, stratified random sampling will be applied.

Sampling Methodology

The project will target households with similar characteristics across various districts in South Africa. Because of this relative homogeneity, **simple random sampling** will be appropriate for survey design. The entire distribution will be grouped by batch, with each batch being subject to a survey to determine its sample size.

When determining sampling strata, the project team will consider regional differences and possible demographic or behavioral variation among the beneficiaries.

Sample Size Calculation

The sample size will be calculated using a 90% confidence level and 10% margin of error, as recommended by established methodologies. It will be based on the total number of distributed ICS units. To enhance reliability, the following sources will be used to inform parameter estimates:

1. **Previous Research** – Data from earlier projects and studies.

2. **Pilot Study** – A preliminary round of data collection may be conducted to obtain baseline estimates.

3. **Expert Judgement** – Where data is not available, informed assumptions will be made by the research team.

Oversampling and Field Verification

To account for potential non-responses, the PP will survey 20% more households than the calculated minimum sample size. Field teams will visit all selected households and inspect the identified sample stoves. Data gathered from these visits will be logged in a standardized monitoring survey sheet, designed specifically for this project.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

This section will be Updated Later

Data / Parameter	

Data unit	
Description	
Value applied:	
Comments	

7.2 Baseline Emissions

This section will be Updated Later

7.3 Project Emissions

This section will be Updated Later

7.4 Leakage Emissions

This section will be Updated Later

7.5 GHG Emission Reductions and Carbon Dioxide Removals

This Section will be Updated Later

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Removal VCUs (tCO_2e)	Total VCUs (tCO_2e)
DD-MMM-YYYY to 31-Dec-YYYY	*Example:* *50,000*	*Example:* *20,000*	*Example:* *10,000*	*Example:* *10,000*	*Example:* *10,000*	*Example:* *20,000*
01-Jan-YYYY to 31-Dec-YYYY						
01-Jan-YYYY to DD-MMM-YYYY						
…						
Total						

State the non-permanence risk rating (%)		
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes	☐ No
For ARR and IFM projects with harvesting, state, in $tCO_2e$the Long-term Average (LTA).		
Has the LTA been updated based on monitored data, if applicable?	☐ Yes If no, provide justification.	☐ No
State, in tCO_2e, the expected total GHG benefit to date		
If a loss occurred (including a loss event or reversal), state the amount of tCO_2e lost:		

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Buffer pool allocation (tCO_2e)	Reductions VCUs (tCO_2e)	Removals VCUs (tCO_2e)	Total VCU issuance (tCO_2e)
DD-MMM-YYYY to 31-Dec-YYYY	*Example:* *50,000*	*Example:* *20,000*	*Example:* *10,000*	*Example:* *4,000*	*Example:* *8,000*	*Example:* *8,000*	*Example:* *16,000*
01-Jan-YYYY to 31-Dec-YYYY							
01-Jan-YYYY to DD-MMM-YYYY							
Total							

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
DD-MMM-YYYY to 31-Dec-YYYY				
01-Jan-YYYY to 31-Dec-YYYY				

…				
01-Jan-YYYY to DD-MMM-YYYY				
Total				

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

N/A

APPENDIX X: <TITLE OF APPENDIX>

N/A